STA355F: Theory of Statistical Practice **Fall 2026**

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1 M-estimation continued

For a correctly specified identifiable model, the population negative log-likelihood has a unique minimizer at the true parameter, provided the expectations are finite. Indeed, its difference from the value at the truth is the nonnegative Kullback–Leibler divergence.

Non-example: Gaussian mixture model.

$$X_1, X_2, \cdots, X_n \overset{\text{i.i.d.}}{\sim} \frac{1}{2}\mathcal{N}(\mu, 1) + \frac{1}{2}\mathcal{N}(-\mu, 1),$$

for some $\mu \in \mathbb{R}$. By flipping the sign of μ, we get the same distribution. So the model is not identifiable, and the population minimizer is not unique.

In such a case, we may consider estimation up to equivalence class, e.g., we can restrict the parameter space to $\Theta = [0, \infty)$, and then the population minimizer is unique.

On the other hand, if identifiability does hold, we have

Corollary 6.1. *Suppose $X_i \overset{\text{i.i.d.}}{\sim} p_{\theta^*}$ with $\theta^* \in \Theta$, and the dominated parametric model is identifiable. If Θ is compact, $p_\theta(x) > 0$ on a common support and is continuous in θ there, and $\mathbb{E}_{\theta^*}[\sup_{\theta\in\Theta} |\log p_\theta(X)|] < \infty$, then any measurable global maximizer $\widehat{\theta}_n^{\text{MLE}}$ is consistent:*

$$\widehat{\theta}_n^{\text{MLE}} \xrightarrow{p} \theta^* \quad \text{as } n \to \infty.$$

2 Z-estimation and asymptotic normality

Consistency of Z-estimators can be proved under suitable identification and uniform convergence conditions, but it does not follow from existence of roots alone. For M-estimation, compactness and continuity guarantee existence of a minimizer. For a general estimating equation, a solution may not exist, and even if it exists, it may not be unique.

So for the Z-estimation theory, we start with the assumption that there exists a solution, and we assume it is already a consistent estimator. Locally, under certain regularity conditions, we can show asymptotic normality of the estimator. This is a local theory, and it does not guarantee existence or uniqueness of the solution.

Problem setup: Write $H_n(\theta) = n^{-1}\sum_{i=1}^n h(\theta, X_i)$ and $H(\theta) = \mathbb{E}[h(\theta, X)]$, where $X_1, \ldots, X_n$ are i.i.d. We use the following sufficient local conditions.

1. The parameter θ^* is an interior point of $\Theta \subseteq \mathbb{R}^d$, $H(\theta^*) = 0$, and $\mathbb{E}[\|h(\theta^*, X)\|_2^2] < \infty$.
2. The function $h(\theta, x)$ is measurable in x. On an open neighborhood of a closed ball U of fixed radius $r > 0$ centered at θ^*, with $U \subseteq \text{int}(\Theta)$, it is continuously differentiable in θ for almost every x. Its derivative is measurable and satisfies

$$\mathbb{E}\Big[\sup_{\theta\in U} \|\nabla_\theta h(\theta, X)\|_{\text{op}}\Big] < \infty.$$

3. The matrix $\nabla_\theta H(\theta^*) = \mathbb{E}[\nabla_\theta h(\theta^*, X)]$ is invertible.

4. A measurable Θ-valued selected root $\widehat{\theta}_n$ satisfies $\mathbb{P}(H_n(\widehat{\theta}_n) = 0) \to 1$ and $\widehat{\theta}_n \xrightarrow{p} \theta^*$.

The last item assumes consistency of the selected root separately. Existence of some root in every neighborhood of θ^* does not ensure that an arbitrary selected root is consistent. These assumptions do not assert global existence or uniqueness of roots.

Theorem 6.1. *Under conditions 1–4, we have*

$$\sqrt{n}(\widehat{\theta}_n - \theta^*) \xrightarrow{d} \mathcal{N}\big(0, \nabla_\theta H(\theta^*)^{-1} \Sigma^* (\nabla_\theta H(\theta^*))^{-\top}\big),$$

where $\Sigma^* = \mathbb{E}[h(\theta^*, X) h(\theta^*, X)^\top]$.

Proof: By the derivative-envelope condition and the uniform law of large numbers on the compact ball U,

$$\sup_{\theta \in U} \|\nabla_\theta H_n(\theta) - \nabla_\theta H(\theta)\|_{\mathrm{op}} \xrightarrow{p} 0.$$

The same envelope justifies differentiating H under the expectation and implies continuity of $\nabla_\theta H$. On the event that $\widehat{\theta}_n \in U$ is a root, Taylor's formula in integral form gives

$$\begin{aligned} 0 &= H_n(\theta^*) + \overline{J}_n(\widehat{\theta}_n - \theta^*), \\ \overline{J}_n &= \int_0^1 \nabla_\theta H_n\big(\theta^* + t(\widehat{\theta}_n - \theta^*)\big)\, dt. \end{aligned}$$

Set $\overline{J}_n = \nabla_\theta H(\theta^*)$ off this event. Consistency and the preceding uniform convergence imply $\overline{J}_n \xrightarrow{p} \nabla_\theta H(\theta^*)$. The event just used has probability tending to one, and $\overline{J}_n$ is invertible with probability tending to one. Meanwhile, the multivariate central limit theorem gives

$$\sqrt{n} H_n(\theta^*) \xrightarrow{d} \mathcal{N}(0, \Sigma^*).$$

Consequently,

$$\begin{aligned} \sqrt{n}(\widehat{\theta}_n - \theta^*) &= -\overline{J}_n^{-1} \sqrt{n} H_n(\theta^*) \\ &= -\nabla_\theta H(\theta^*)^{-1} \sqrt{n} H_n(\theta^*) + o_p(1), \end{aligned}$$

where the first equality holds with probability tending to one. Slutsky's theorem proves the result. This argument establishes the $\sqrt{n}$ rate directly, without assuming in advance that a Taylor remainder is $o_p(n^{-1/2})$.

Statistical inference based on this result: Σ^* and $\nabla_\theta H(\theta^*)$ are unknown, but we can estimate them based on data.

$$\begin{aligned} \widehat{\Sigma}_n &= \frac{1}{n} \sum_{i=1}^n h(\widehat{\theta}_n, X_i) h(\widehat{\theta}_n, X_i)^\top, \\ \widehat{J}_n &= \frac{1}{n} \sum_{i=1}^n \nabla_\theta h(\widehat{\theta}_n, X_i). \end{aligned}$$

For covariance estimation, additionally assume

$$\mathbb{E}\Big[\sup_{\theta\in U}\|h(\theta,X)\|_2^2\Big] < \infty.$$

Continuity and the uniform law of large numbers then give $\widehat{\Sigma}_n \xrightarrow{p} \Sigma^*$ and $\widehat{J}_n \xrightarrow{p} \nabla_\theta H(\theta^*)$. If also Σ^* is positive definite, the estimated sandwich covariance is positive definite with probability tending to one, and

$$\sqrt{n}\Big(\widehat{J}_n^{-1}\widehat{\Sigma}_n\widehat{J}_n^{-\top}\Big)^{-1/2}(\widehat{\theta}_n - \theta^*) \xrightarrow{d} \mathcal{N}(0, I_d), \quad \text{as } n\to\infty.$$

We can then build confidence sets and perform hypothesis testing based on this result.

Remark: this is just one method for inference. We can also use bootstrap to estimate the asymptotic covariance matrix, or we can use bootstrap to directly estimate the distribution of $\widehat{\theta}_n$.

Implication to M-estimation: An interior minimizer of a continuously differentiable empirical loss solves its gradient estimating equation. Thus we can apply the Z-estimation result to a consistent M-estimator when the following local conditions hold.

Verifying the problem setup:

- Population estimating equation: an interior population minimizer θ^* satisfies $\nabla F(\theta^*) = 0$, with differentiation under the expectation justified by the local envelope conditions.
- If a measurable empirical minimizer $\widehat{\theta}_n$ is consistent and θ^* is interior, then $\widehat{\theta}_n$ is interior with probability tending to one and satisfies the estimating equation $\nabla F_n(\widehat{\theta}_n) = 0$.
- For $h(\theta, X) = \nabla_\theta f(\theta, X)$, require f to be twice continuously differentiable locally, with

  $$\mathbb{E}[\|\nabla_\theta f(\theta^*, X)\|_2^2] < \infty,$$

  and an integrable local envelope for $\|\nabla_\theta^2 f(\theta, X)\|_{\text{op}}$, as in conditions 1–3 of Theorem 6.1.
- Non-singularity of $\nabla_\theta H(\theta^*) = \nabla^2 F(\theta^*)$. Since θ^* is the minimizer of F, this condition is equivalent to the Hessian of F being positive definite at θ^*.

In such a case, we have

$$\sqrt{n}(\widehat{\theta}_n - \theta^*) \xrightarrow{d} \mathcal{N}(0, \nabla^2 F(\theta^*)^{-1}\Sigma^*\nabla^2 F(\theta^*)^{-1}), \quad \text{as } n\to\infty,$$

where $\Sigma^* = \mathbb{E}[\nabla_\theta f(\theta^*, X)\nabla_\theta f(\theta^*, X)^\top]$.

Specialization to MLE: Assume the model is correctly specified and the preceding local M-estimation conditions hold for $f(\theta, x) = -\log p_\theta(x)$. For the information identity, also assume that the densities are positive on a common parameter-independent support and that the normalizing integral $\int p_\theta(x)\,dx = 1$ can be differentiated under the integral sign twice in a neighborhood of the true parameter. Then we can compute

$$\Sigma^* = \mathbb{E}_{\theta^*}[\nabla_\theta \log p_{\theta^*}(X)\nabla_\theta \log p_{\theta^*}(X)^\top], \qquad \nabla^2 F(\theta^*) = -\mathbb{E}_{\theta^*}[\nabla_\theta^2 \log p_{\theta^*}(X)].$$

Theorem 6.2 (Fisher's identity)**.** *Under the stated common-support and differentiation conditions, $\Sigma^* = \nabla^2 F(\theta^*)$. This is the Fisher information matrix for one observation, denoted as $I(\theta^*)$.*

Proof: first, for the score function $\nabla \log p_\theta(X)$, we have

$$\mathbb{E}_\theta[\nabla_\theta \log p_\theta(X)] = \int p_\theta(x) \nabla_\theta \log p_\theta(x) dx = \int \nabla_\theta p_\theta(x) dx = \nabla_\theta \int p_\theta(x) dx = 0. \tag{6.1}$$

We can differentiate both sides of (6.1), for the i, j-th entry of the matrix, we have

$$\int \partial_{\theta_i} \partial_{\theta_j} \log p_\theta(x) \cdot p_\theta(x) dx + \int \partial_{\theta_j} \log p_\theta(x) \cdot \partial_{\theta_i} \log p_\theta(x) \cdot p_\theta(x) dx = 0.$$

By collecting these entries into a matrix, we have

$$\mathbb{E}_\theta[\nabla_\theta^2 \log p_\theta(X)] + \mathbb{E}_\theta[\nabla_\theta \log p_\theta(X) \nabla_\theta \log p_\theta(X)^\top] = 0.$$

Using the Fisher's identity in combination with the asymptotic normality result, we have

$$\sqrt{n}(\widehat{\theta}_n - \theta^*) \xrightarrow{d} \mathcal{N}(0, I(\theta^*)^{-1}), \quad \text{as } n \to \infty,$$

for MLE, under the same regularity conditions as above.

Remarks:

- Intuitively, the Fisher information matrix $I(\theta^*)$ measures how much information the data contains about the unknown parameter θ^*. The larger the Fisher information, the smaller the asymptotic variance of the MLE, and the more accurate the estimation.
- There are cases where the Fisher information matrix is singular, and in such a case, the asymptotic normality may not hold. But MLE may still be consistent, and it may converge at a slower rate than $1/\sqrt{n}$.
- For inference, we just need to use plug-in method to estimate the Fisher information matrix.

2.1 Cramér–Rao lower bound

Fisher information is important because it captures the fundamental possibility of estimation. One example is the Cramér–Rao lower bound, which gives a lower bound on the variance of any unbiased estimator.

Idea: we want to find the best one among all possible estimators. This is generally not possible (decision theory), but we can restrict to unbiased estimators. Then we can show that the variance of any unbiased estimator is lower bounded by the inverse of Fisher information matrix.

Theorem 6.3 (Cramér–Rao lower bound)**.** *Suppose the model satisfies the common-support and differentiation conditions above, $I(\theta)$ is positive definite, and $\widehat{\theta}_n$ is unbiased throughout a neighborhood of θ, with finite covariance. Assume also that differentiation can pass under its expectation. For n i.i.d. observations, we have*

$$\mathrm{Cov}_\theta(\widehat{\theta}_n) \succeq \frac{1}{n} I(\theta)^{-1},$$

where $A \succeq B$ means that $A - B$ is positive semi-definite.

Proof (for simplicity, we consider the one-dimensional case): by unbiasedness, we have

$$\theta = \int \widehat{\theta}_n(x_1, x_2, \ldots, x_n) \prod_{i=1}^n p_\theta(x_i) dx_1 dx_2 \cdots dx_n, \tag{6.2}$$

for θ throughout the neighborhood on which unbiasedness is assumed. Differentiating both sides of (6.2) with respect to θ, we have

$$1 = \int \widehat{\theta}_n(x_1, x_2, \ldots, x_n) \cdot \sum_{i=1}^n \partial_\theta \log p_\theta(x_i) \cdot \prod_{i=1}^n p_\theta(x_i) dx_1 dx_2 \cdots dx_n \tag{6.3}$$

On the other hand, we have score identity (6.1):

$$0 = \int \sum_{i=1}^n \partial_\theta \log p_\theta(x_i) \cdot \prod_{i=1}^n p_\theta(x_i) dx_1 dx_2 \cdots dx_n. \tag{6.4}$$

(6.3) $-\ \theta\times$ (6.4), we have

$$1 = \int (\widehat{\theta}_n - \theta) \cdot \sum_{i=1}^n \partial_\theta \log p_\theta(x_i) \cdot \prod_{i=1}^n p_\theta(x_i) dx_1 dx_2 \cdots dx_n = \mathbb{E}_\theta\Big[\underbrace{(\widehat{\theta}_n - \theta)}_{\text{estimation error}} \cdot \sum_{i=1}^n \partial_\theta \log p_\theta(X_i)\Big]$$

$$\le \sqrt{\mathrm{var}_\theta(\widehat{\theta}_n - \theta)} \cdot \sqrt{\mathrm{var}_\theta\Big(\sum_{i=1}^n \partial_\theta \log p_\theta(X_i)\Big)} = \sqrt{\mathrm{var}_\theta(\widehat{\theta}_n)} \cdot \sqrt{nI(\theta)},$$

Squaring and rearranging gives $\mathrm{var}_\theta(\widehat{\theta}_n) \ge 1/(nI(\theta))$. In dimension d, let $S_n = \sum_{i=1}^n \nabla_\theta \log p_\theta(X_i)$. Differentiated unbiasedness gives $\mathrm{Cov}_\theta(\widehat{\theta}_n, S_n) = I_d$, whereas $\mathrm{Cov}_\theta(S_n) = nI(\theta)$. Therefore

$$0 \preceq \mathrm{Cov}_\theta\Big(\widehat{\theta}_n - \theta - (nI(\theta))^{-1} S_n\Big) = \mathrm{Cov}_\theta(\widehat{\theta}_n) - \frac{1}{n} I(\theta)^{-1},$$

which proves the matrix bound as well.

Remarks:

- Unbiasedness is very restrictive, and MLE itself is not unbiased in general. So this lower bound itself does not prove optimality of MLE.
- Fisher information is also useful without finite-sample unbiasedness. In regular models, the convolution theorem says that a regular estimator's limiting law is a convolution of $\mathcal{N}(0, I(\theta^*)^{-1})$ with another probability law. An efficient MLE attains the Gaussian part without extra noise. The convolution and local asymptotic minimax theorems formalize this asymptotic optimality; their hypotheses and proofs are beyond the scope of this course.

2.2 Additional topics

Computational algorithms for M and Z estimation:

- In some cases, we can solve them with very high precision.
- In some cases, we want to build statistical theory for approximate solutions that are computed from algorithms.

Here I present two classical examples.

Newton's method: Suppose we want to solve $H_n(\theta) = 0$. We can use Newton's method to iteratively update the solution:

$$\theta^{(t+1)} = \theta^{(t)} - \nabla_\theta H_n(\theta^{(t)})^{-1} H_n(\theta^{(t)}).$$

(under certain conditions, this converges at super-linear rate locally around the true solution).

On the other hand, for statistical analysis, sometimes just one-step Newton's method is enough.

Theorem 6.4. *Assume conditions 1–3 of Theorem 6.1, and let a measurable Θ-valued starting estimator $\widetilde{\theta}_n$ satisfy $\left\|\widetilde{\theta}_n - \theta^*\right\|_2 = O_p(n^{-1/2})$. On the event where $\widetilde{\theta}_n \in U$ and the derivative is invertible, define the one-step estimator by*

$$\widehat{\theta}_n = \widetilde{\theta}_n - \nabla_\theta H_n(\widetilde{\theta}_n)^{-1} H_n(\widetilde{\theta}_n).$$

Define it arbitrarily on the complement, whose probability tends to zero. Then we have

$$\sqrt{n}(\widehat{\theta}_n - \theta^*) \xrightarrow{d} \mathcal{N}(0, \nabla_\theta H(\theta^*)^{-1} \Sigma^* (\nabla_\theta H(\theta^*))^{-\top}), \quad \text{as } n \to \infty.$$

Operationally, this allows us to find a computable estimator that is asymptotically equivalent to the Z-estimator.

Proof idea:

$$\begin{aligned}
&\widetilde{\theta}_n - \nabla_\theta H_n(\widetilde{\theta}_n)^{-1} H_n(\widetilde{\theta}_n) \\
&= \widetilde{\theta}_n - \nabla_\theta H_n(\widetilde{\theta}_n)^{-1} \Big\{ H_n(\theta^*) + \nabla H_n(\theta^*)(\widetilde{\theta}_n - \theta^*) + o_p(\left\|\widetilde{\theta}_n - \theta^*\right\|_2) \Big\}
\end{aligned}$$

The local uniform derivative convergence proved above gives $\nabla H_n(\widetilde{\theta}_n) \xrightarrow{p} \nabla H(\theta^*)$ and justifies the Taylor remainder in this display. Hence the terms in blue cancel up to $o_p(n^{-1/2})$. This gives

$$\widehat{\theta}_n = \theta^* - \nabla H_n(\theta^*)^{-1} H_n(\theta^*) + o_p(1/\sqrt{n}),$$

which proves the asymptotic normality of $\widehat{\theta}_n$.

Stochastic gradient descent:

$$\theta_{t+1} = \theta_t - \eta_t \nabla_\theta f(\theta_t, X_{t+1}),$$

for $t = 0, 1, \cdots, n-1$.

Polyak–Juditsky–Ruppert averaging: $\widehat{\theta}_n = \frac{1}{n}\sum_{t=1}^n \theta_t$.

Under certain condition, we can show that $\widehat{\theta}_n$ is asymptotically normal and achieves the same asymptotic covariance as the M-estimator.